

## **Marine Bioinformatics Hackathon: Biodiversity, Genomics, and Nextflow Pipelines.**

**Venue: Nile University 29-31 October 2025**

| <b>Day 1 (29- 10 -2025)</b> |                                                                                                                                       |
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| <b>Timeline</b>             | <b>Lectures and practical sessions</b>                                                                                                |
| <b>9:00 – 9:30</b>          | Registration                                                                                                                          |
| <b>10:00 - 10:15</b>        | Welcome Note (Dr. Asmaa Abushady) & AfricaBP Introductory Session- Room 367, UB1                                                      |
| <b>10:15 – 11:15</b>        | Lec1. Biodiversity Loss in Marine Biology (Dr. Sameh) Room 367, UB1                                                                   |
| <b>11:15 – 12:15</b>        | Lec2. Next Generation technologies: With a Deep dive in Marine Biology 367, UB1                                                       |
| <b>12:30 – 13:00</b>        | Coffee break - Appetisers                                                                                                             |
| <b>13:00 – 14:00</b>        | Lec3. Applications of Metagenomics in Marine Drug Discovery (Dr. Sameh) - Room 367, UB1                                               |
| <b>14:00- 15:00</b>         | Practical Session 1: Quick tutorial on Galaxy platform and illustration for the selected pipeline (Shaimaa and Assem) - Room 367, UB1 |
| <b>15:00- 16:30</b>         | Practical Session 2: Getting data, QC and Sequence alignment (Shaimaa and Assem) - Room 367, UB1                                      |

| Day 2 (30 - 10 -2025) |                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Timeline              | Lectures and practical sessions                                                                                           |
| 9:00 – 10:00          | Lec1.Integrated metagenome-metabolome analysis reveals coral responses to ocean warming (Dr. Amin Roshdy) - Room 367, UB1 |
| 10:00 – 10:30         | Practical Session3. Extraction of Taxonomic Information (Shaimaa and Assem) - Room 53, UB1                                |
| 10:30 – 11:30         | Practical Session4. Visualization (Shaimaa and Assem) - Room 127, UB1                                                     |
| 11:30 - 12:30         | Coffee break - Appetizers                                                                                                 |
| 12:30 – 14:00         | Lec2. Biodiversity Genomics in the '-Omics' Era (Dr.Abdollah Sharaf) Dr.Abdollah Sharaf - Room 127 , UB1                  |
| 14:00- 15:00          | Hackathon : Groups division and Projects assigning 367, UB1                                                               |
| 15:00- 16:30          | Hackathon : Working on group projects 367, UB1                                                                            |

| <b>Day 3 (31- 10-2025)</b> |                                                              |
|----------------------------|--------------------------------------------------------------|
| <b>Timeline</b>            | <b>Lectures and practical sessions</b>                       |
| <b>9:00 – 10:30</b>        | Workflow management with Nextflow                            |
| <b>10:30 – 11:00</b>       | Coffee break - Appetizers                                    |
| <b>11:00 – 12:00</b>       | Introduction to nf-core                                      |
| <b>12:00 - 12:30</b>       | Prima Project activities (Dr.Asmaa Abushady) - Room 367, UB1 |
| <b>12:30 - 15:30</b>       | Project discussions and honoring best group                  |
| <b>15:30 – 16:00</b>       | Closing Ceremony                                             |

Visit the links below to access workshop practical materials

[https://drive.google.com/drive/folders/1fCm6gp2tY-7LLAp-Pl-n5aMfiCd85pJY?usp=share\\_link](https://drive.google.com/drive/folders/1fCm6gp2tY-7LLAp-Pl-n5aMfiCd85pJY?usp=share_link)

Please Note:

- Lectures' materials will be uploaded by the end of the workshop